WEB-BASED APPLICATION DEVELOPMENT

(CSE3023)

<u>Lab Report 2</u>

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Group: K1

Programme: Bachelor of Computer Science (Software Engineering) With Honors

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2.9 EXERCISE

CODE:

Product.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.lab.model;


public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    public Product(){}

    public Product (int id, String name, String category, double price, int quantity){
        this.id=id;
        this.name=name;
        this.category =category;
        this.price=price;
        this.quantity = quantity;
    }


// Constructor tanpa ID (Untuk Insert)
public Product(String name, String category, double price, int quantity) {
this.name = name;
this.category = category;
this.price = price;
this.quantity=quantity;
}

// Getter dan Setter
public int getId() {
return id;
}
```

```
38  public void setId(int id) {
39  this.id = id;
40  }
41  public String getName() {
42  return name;
43  }
44  public void setName(String name) {
45  this.name = name;
46  }
47  public String getCategory() {
48  return category;
49  }
50  public void setCategory(String category) {
51  this.category = category;
52  }
53  public double getPrice() {
54  return price;
55  }
56  public void setPrice(double price) {
57  this.price = price;
58  }
59  public int getQuantity(){
60      return quantity;
61  }
62  public void setQuantity(int quantity){
63      this.quantity=quantity;
64  }
65  }
```

ProductDAO.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt t
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit thi
 */
package com.lab.dao;


import com.lab.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;


public class ProductDAO {
private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
private String jdbcUsername = "root";
private String jdbcPassword = "admin";

// Method untuk mendapatkan sambungan Database
protected Connection getConnection() {
    Connection connection = null;
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        System.out.println("Connecting to DB...");

        connection = DriverManager.getConnection(
            jdbcURL, jdbcUsername, jdbcPassword);

        System.out.println("✅ SUCCESS CONNECT");
```

```java
    } catch (Exception e) {
        System.out.println("❌ CONNECTION FAILED");
        e.printStackTrace();
    }
    return connection;
}


// CREATE: Masukkan pengguna baharu
public void insertProduct(Product product) throws SQLException {
String sql = "INSERT INTO products (name, category,price,quantity) VALUES (?, ?, ?,?)";
try (Connection conn = getConnection()){

        if (conn == null) {
            System.out.println("❌ Conn is NULL - stop here");
            return;
        }
        PreparedStatement ps = conn.prepareStatement(sql) ;
        ps.setString(1, product.getName());
        ps.setString(2, product.getCategory());
        ps.setDouble(3, product.getPrice());
        ps.setInt(4, product.getQuantity());
        ps.executeUpdate();
} catch (SQLException e) {
e.printStackTrace();
}
}



public List<Product> selectAllProducts() {
        List<Product> list = new ArrayList<>();
        try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement("SELECT * FROM products")) {
            ResultSet rs = ps.executeQuery();
            while (rs.next()) {
```

```java
                list.add(new Product(rs.getInt("id"), rs.getString("name"),
                        rs.getString("category"), rs.getDouble("price"), rs.getInt("quantity")));
            }
        } catch (SQLException e) { e.printStackTrace(); }
        return list;
    }

    public Product selectProduct(int id) {
        Product product = null;
        try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement("SELECT * FROM products WHERE id = ?")) {
            ps.setInt(1, id);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                product = new Product(id, rs.getString("name"), rs.getString("category"),
                                      rs.getDouble("price"), rs.getInt("quantity"));
            }
        } catch (SQLException e) { e.printStackTrace(); }
        return product;
    }

    public boolean updateProduct(Product p) {
        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement("UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?")) {
            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());
            ps.setInt(5, p.getId());
            return ps.executeUpdate() > 0;
        } catch (SQLException e) { return false; }
    }

    public boolean deleteProduct(int id) {
        try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement("DELETE FROM products WHERE id = ?")) {
            ps.setInt(1, id);
            return ps.executeUpdate() > 0;
```

```java
            } catch (SQLException e) { return false; }
        }
    }
```

InsertProductServlet.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this licen
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */
package com.lab.controller;


import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;


public class InsertProductServlet extends HttpServlet {

    private ProductDAO productDAO = new ProductDAO();



    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
```

```java
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet InsertProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet InsertProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
```

```java
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {


        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product newProduct = new Product(name, category, price, quantity);
        try {
            productDAO.insertProduct(newProduct);
        } catch (SQLException ex) {
            Logger.getLogger(InsertProductServlet.class.getName()).log(Level.SEVERE, null, ex);
        }

        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

ViewProductServlet.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;


public class ViewProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }
    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
```

```java
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet ViewProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet ViewProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        List<Product> products = productDAO.selectAllProducts();

        out.println("<h2>Product Inventory</h2>");
        out.println("<table border='1'><tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actions</th></tr>");

        for (Product p : products){
            out.println("<tr>");
```

```
74              out.println("<td>" + p.getId() + "</td>");
75              out.println("<td>" + p.getName() + "</td>");
76              out.println("<td>" + p.getCategory() + "</td>");
77              out.println("<td>" + p.getPrice() + "</td>");
78              out.println("<td>" + p.getQuantity()+ "</td>");
79              out.println("<td>"
80          + "<a href='UpdateProductServlet?id=" + p.getId() + "'>Edit</a> | "
81          + "<a href='DeleteProductServlet?id=" + p.getId() + "' onclick='return confirm(\"Delete this product?\")'>Delete</a>"
82          + "</td>");
83              out.println("</tr>");
84
85          }
86
87          out.println("</table>");
88          out.println("<br><a href = 'add_product.html'>Add New Product</a>");
89      }
90
91      /**
92       * Handles the HTTP <code>POST</code> method.
93       *
94       * @param request servlet request
95       * @param response servlet response
96       * @throws ServletException if a servlet-specific error occurs
97       * @throws IOException if an I/O error occurs
98       */
99      @Override
100     protected void doPost(HttpServletRequest request, HttpServletResponse response)
101             throws ServletException, IOException {
102         processRequest(request, response);
103     }
104
105     /**
106      * Returns a short description of the servlet.
107      *
108      * @return a String containing servlet description
109      */
```

```
110         @Override
111         public String getServletInfo() {
112             return "Short description";
113         }// </editor-fold>
114
115     }
116
```

UpdateProductServlet.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this li
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */
package com.lab.controller;


import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;


public class UpdateProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
```

```java
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet UpdateProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet UpdateProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        int id = Integer.parseInt(request.getParameter("id"));

        Product existingProduct = productDAO.selectProduct(id);

        if(existingProduct != null){
            out.println("<h2>Edit Product</h2>");
            out.println ("<form action = 'UpdateProductServlet' method='post'>");
            out.println ("<input type = 'hidden' name='id' value='" + existingProduct.getId() + "'>");
            out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "'><br><br>");
```

```
        out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "'><br><br>");
        out.println("Price: <input type='number' step='0.01' name='price' value='" + existingProduct.getPrice() + "'><br><br>");
        out.println("Quantity: <input type='number' name='quantity' value='" + existingProduct.getQuantity() + "'><br><br>");
        out.println("<input type='submit' value='Update Product'>");
        out.println("</form>");
    }
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity");

    Product updatedProduct = new Product(id, name, category, price, quantity);
    productEAO.updateProduct(updatedProduct);
    response.sendRedirect("ViewProductServlet");
}

/**
 * Returns a short description of the servlet.
 *
 * @return a String containing servlet description
 */
```

```
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

DeleteProductServlet.java

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this template
 */
package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author Dell
 */
public class DeleteProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init(){
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
```

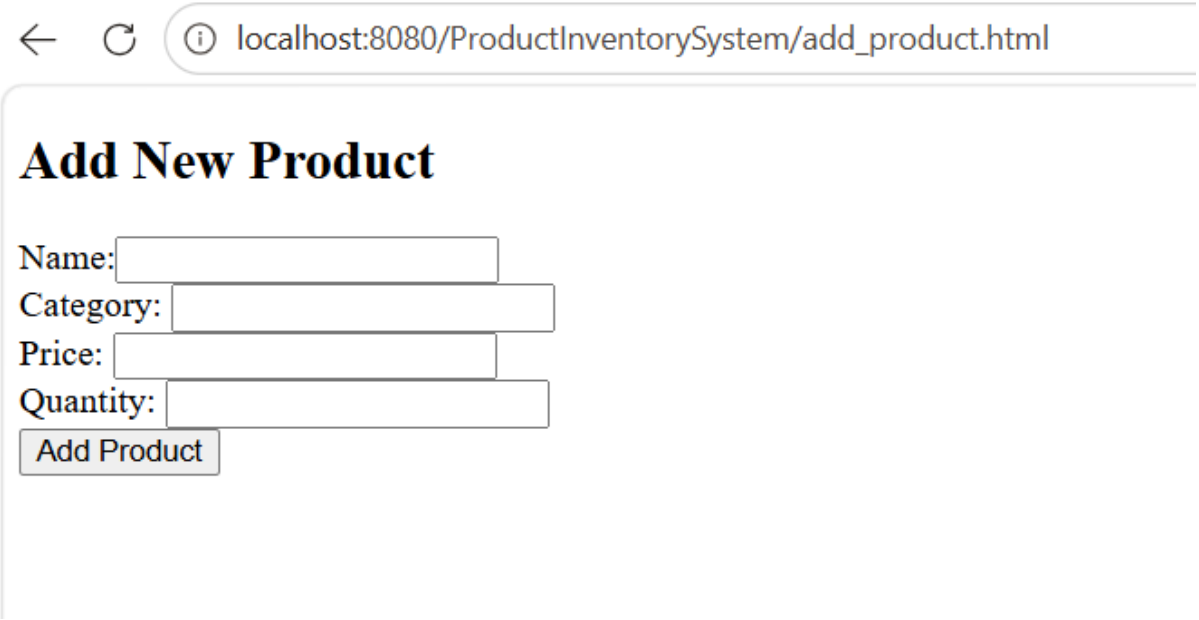
```java
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet DeleteProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet DeleteProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        productDAO.deleteProduct(id);
        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
```

```
74          * @param response servlet response
75          * @throws ServletException if a servlet-specific error occurs
76          * @throws IOException if an I/O error occurs
77          */
78         @Override
80         protected void doPost(HttpServletRequest request, HttpServletResponse response)
                 throws ServletException, IOException {
81             processRequest(request, response);
82         }
83
84         /**
85          * Returns a short description of the servlet.
86          *
87          * @return a String containing servlet description
88          */
89         @Override
91         public String getServletInfo() {
92             return "Short description";
92         }// </editor-fold>
93
94     }
95
```

add_product.html

```html
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Html.html to edit this template
-->
<html>
    <head>
        <title>Add Product</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Add New Product</h2><!-- comment -->
        <form action ="InsertProductServlet" method ="POST">
            Name:<input type="text" name="name"><br>
            Category: <input type="text" name="category"><br>
            Price: <input type="number" step="0.01" name="price"><br>
            Quantity: <input type="number" name="quantity"><br>
            <input type="submit" value="Add Product">
        </form>
    </body>
</html>
```

OUTPUT:


localhost:8080/ProductInventorySystem/add_product.html
Add New Product
Name:
Category:
Price:
Quantity:
Add Product

localhost:8080/ProductInventorySystem/add_product.html
Add New Product
Name: iphone 14
Category: mobile phone
Price: 2400
Quantity: 3
Add Product

Product Inventory

ID	Name	Category	Price	Quantity	Actions
7	iphone 14	mobile phone	2400.0	3	Edit | Delete

Add New Product

Product Inventory

ID	Name	Category	Price	Quantity	Actions
7	iphone 14	mobile phone	2400.0	3	Edit | Delete
8	ipad 10th generation	tablet	1899.0	5	Edit | Delete

Add New Product

Product Inventory

ID	Name	Category	Price	Quantity	Actions
7	iphone 14	mobile phone	2400.0	7	Edit | Delete
8	ipad 10th generation	tablet	1899.0	5	Edit | Delete

Add New Product

Product Inventory

ID	Name	Category	Price	Quantity	Actions
8	ipad 10th generation	tablet	1899.0	5	Edit | Delete

Add New Product